MCAST

ASSESSMENT AND INTERNAL VERIFICATION FRONT SHEET (Individual Criteria)

(Note : This version is to be used for an assignment brief issued to students via Classter)

Course Title	**Diploma in ICT/ iGaming (MQF Level 3)**		**Lecturer Name & Surname**		
Unit Number & Title	**ITDBS-306-1601 - Database Design and Development**				
Assignment Number, Title / Type	**Databases Synoptic Assignment**				
Date Set	**4th June 2021**	**Deadline Date**	**14th June 2021**		
Student Name		**ID Number**		**Class / Group**	

Assessment Criteria	Maximum Mark
K-7: State properly basic queries.	6
K-8: Define the different kinds of reports.	4
K-9: By referring to an implemented database, explain common errors which could occur.	4
K-10: Describe how to create proper forms using a DBMS.	6
U-2: Discuss relationships present in a database.	6
U-3: Summarise use of different data retrieval methods used in a given scenario.	6
U-4: Show the difference between simple queries and compound queries applied in a given scenario.	10
U-5: Give an example of basic data validation.	10
A-1: Produce a structured database for a given scenario.	24
A-2: Apply compound queries for a database.	10
A-3: Produce forms and views using suitable data validation including output reports.	14
Total Mark	100

Notes to Students:

- This assignment brief has been approved and released by the Internal Verifier through Classter.
- Assessment marks and feedback by the lecturer will be available online via Classter (http://mcast.classter.com) following release by the Internal Verifier
- Students submitting their assignment on Moodle/Unicheck will be requested to confirm online the following statements:

Student's declaration prior to handing-in of assignment

- ❖ I certify that the work submitted for this assignment is my own and that I have read and understood the respective Plagiarism Policy

Student's declaration on assessment special arrangements

- ❖ I certify that adequate support was given to me during the assignment through the Institute and/or the Inclusive Education Unit.
- ❖ I declare that I refused the special support offered by the Institute.

ITDBS-306-1601
Database Design and Development

Diploma in Computing

Synoptic

Databases Synoptic Assignment

Assignment Guidelines

Read the following instructions carefully before you start the assignment.
If you have any difficulties, ask your lecturer.

1. This Synoptic assignment is a **Home Assignment** and should be completed and submitted online to your lecturer by **14th June 2021**. The assignment will be issued on 4th June 2021.

2. Fill in the **assignment cover sheet appropriately**. The student's assignment cover sheet should be attached to the project being submitted. Note that assignments handed in without the cover sheet may not be accepted (thus considered as **Not Submitted**).

3. Design and create a front **cover sheet within the documentation** by inserting the following details:

   I. Course Title;
   II. Unit Number and Title;
   III. Assignment Number and Name;
   IV. Your Name and Surname;
   V. Your Group No.

4. Answer all the questions in a separate document or relevant programs.

5. **Plagiarism is strictly prohibited**, so proper references should be used. Likewise, **Copying is strictly prohibited** and will be penalised in line with the College's disciplinary procedures.

6. Clearly highlight the Task/Question Number for all the answers presented.

7. Submit the project through an online tool as directed by your lecturer. **All work produced**, including the document, the database and any other file(s) must also be submitted as per guided by your lecturer.

8. The assignment carries a **total of 100%**

Assignment Scenario

A local gym has requested the design and the development of a database solution for its membership processes. Following initial meetings with the client, the following entities and attributes have been identified:

Trainer: Code; Name; Surname; Mobile_No_; Salary; Date_Employed; Part-Time(Y/N)

Member: ID; Name; Surname; DOB; House_No_; House_Name; Street; Locality; Post_Code; Mobile_No_; Email; Date_First_Joined

Plan: Code; Description; No_of_Days; Cost

Membership: …

Further details and assumptions (Note that you might also be required to add extra fields):

1. The gym started operating on 1st March 2019. No trainer was employed prior that date.
2. Trainer's salaries range between €3,000 and €19,500 per annum.
3. A plan describes the type of subscription available to members. The gym has got the following plans:
    a. One-Day
    b. One-Month
    c. Six-Month
    d. One-Year

4. The membership entity implies a subscription to the gym by a member. During the booking process, each member must select a plan and is also assigned a trainer. All membership lines must have a start date accordingly.

Task 1 – (U2, U5, A1 ***[includes K4, K5, U1]*****)** Database Structure

Part 1:

Design and model the above scenario requirements through a physical Entity-Relationship Diagram (ERD). The diagram should include:

i. All entities being specified;
ii. All respective attributes;
iii. An indication of the attributes which are a Primary Key or a Foreign Key (datatypes are optional);
iv. All the relationships specified along with their cardinality.

The diagram should also be built through a relevant software or online tool such as Diagrams.net (Draw.io) or any other diagramming application of your choice. The ERD must be inserted into the assignment document accordingly. Hand-drawn diagrams are not acceptable. **Note:** where applicable, all files/links created must be submitted as well.

Part 2:

Using the ERD created in *Part 1*, develop a corresponding database through MS Access. The following details must be observed:

i. Create all tables, fields, and data types;
ii. Give a suitable name wherever needed (including a proper Database name);
iii. Select appropriate data types;
iv. Amend the properties respectively;
v. Create the relationships.

Part 3:

In the assignment document, and, where applicable through the Scenario presented in this assignment:

1. Mention the three types of relationships possible in an RDB.
2. Explain each relationship created above by referring to the entities involved and the relationship cardinality. (For example: *"A relationship between Author and Books which implies that 1 Author can write many Books"*).
3. In your own words, explain 2 advantages of using relationships in a database. Ideally, you use examples from the assignment scenario.

Part 4:

Identify 3 fields across all tables and create a different validation rule for each. Take a screenshot of the 3 rules and insert them into your document.

*(**Tip:** You might consider enforcing the dates specified above and/or the salary range validation amongst other options.)*

Part 5:

Populate the created tables with the following:

i. 7 Trainers;
ii. 15 Members;
iii. 4 Plans;
iv. 35 Memberships.

The data must be fictious, but it must make sense (i.e., random-typed characters are not accepted). **Also note** that offensive or vulgar data is strictly not tolerated and will thus invalid all relevant tasks.

Task 2 – (K10, A3) Forms and Reports

Part 1:

The end user must operate the database through Forms not Tables. Thus, you are to create 1 form for each table and name it as follows:

"*<Name of Table>* Input Form"

In total you must therefore create 4 forms using all fields available and the *Columnar* layout. Furthermore, you are to create a step-by-step guide with screenshots to illustrate the process which you went through to create any 1 of the 4 forms.

*(**Note:** Screenshots must be of proper size and must thus provide a readable view. Careless work will be penalised.)*

Part 2:

The client has requested 2 reports as follows:

1. "*All Members Report*" - This report should display all member records sorted according to the first membership date (oldest date first) and then by surname (descending). Use *Justified* layout and *Landscape* orientation.

2. "*Trainers per Employment Type Report*" – This report should display a list of all trainers grouped by their job status (i.e., full-time, or part-time). The list should be sorted by date of employment in ascending order, respectively. Use *Outline* Layout and *Portrait* orientation.

Task 3 – (K7, U4, A2) Queries

Part 1:

The client wants a quick way to get specific information on a regular basis, and so we have suggested the use of Queries. You are to create a step-by-step guide with screenshots to illustrate the process necessary to create any 1 of the following 2 queries (both queries must be created through Query Design):

1. "*One-Year Memberships Query*" – This query will list all yearly memberships. Select just the Start Date, Member ID, and Plan Code, and sort by the Member ID in descending order.

2. "*Highest-Paid Query*" – This query will list all trainers who earn more than €15,000. Select just the Code, Name, Surname and Salary.

*(**Note:** It does not matter if you had created different names for the fields requested.)*

Part 2:

State <u>and explain</u> whether the queries created in *Part 1* are Simple or Compound queries.

Part 3:

Create 2 compound queries of your choice from the database created. Give a suitable name to identify the queries. *(**Hint:** Compound Queries use more than one criteria field. Fields from related tables may also be used.)*

Task 4 – (K8, K9, U3) Other Reporting

Various types of sample reporting will be generated and given to the client. Thus, you need to address the following:

Part 1:

Create Reports based on each and every query built in Task 3. Use the same name and replace the term *'Query'* with *'Report'*. Choose any layout or orientation which you deem appropriate.

Part 2:

Using the scenario and the database implemented, list and explain in about 100 words the common errors which could occur throughout the whole design and development process. You may also refer to examples from the database which you have created.

Part 3:

Create a step-by-step guide with screenshots to illustrate the process required to export data and information in 3 different ways. Use any object which you prefer to demonstrate the process with.

You must also export the following information in the specified format:

1. 'Membership' table as an *Excel file*;
2. 'Plan' table as a *Text file*;
3. 'All Members Report' as a *PDF or XPS file*;

*(**Note:** The three generated files must also be submitted as part of your project.)*

-END-

Assessment Sheet *(to be filled by the assessor)*

Criteria	Tasks	Maximum Mark(s)	Mark(s) Awarded
K-7: State properly basic queries.	Presented a proper step-by-step guide with screenshots to create a query.	6	
K-8: Define the different kinds of reports.	Created 4 reports based on 4 queries.	4	
K-9: By referring to an implemented database, explain common errors which could occur.	Listed and explained the common errors which can occur in a database.	4	
K-10: Describe how to create proper forms using a DBMS.	Presented a proper step-by-step guide with screenshots to create a form.	6	
U-2: Discuss relationships present in a database.	Mentioned the 3 different types of relationships found in an RDB.	1	
	Explained each relationship created, using the entities involved and their cardinality.	3	
	Explained 2 advantages of using relationships in an RDB.	2	
U-3: Summarise use of different data retrieval methods used in a given scenario.	Presented a proper step-by-step guide with screenshots to create an export.	3	
	Exported information as requested in 3 different formats.	3	
U-4: Show the difference between simple queries and compound queries applied in a given scenario.	Created the 2 queries as per specified.	8	
	Stated and explained why the queries are Simple or Compound type.	2	
U-5: Give an example of basic data validation.	Created 3 different but relevant validation rules with appropriate text.	6	
	Inserted screenshots of all validation rules.	4	
A-1: Produce a structured database for a given scenario.	Created ERD as required and inserted diagram in the document submitted.	7	

	Created proper table structures in MS Access.	6	
	Created and enforced all relationships between tables.	6	
	Filled in the data respectively as per guided.	5	
A-2: Apply compound queries for a database.	Created 2 correct compound queries with suitable names.	10	
A-3: Produce forms and views using suitable data validation including output reports.	Created all 4 forms as per requested.	8	
	Created the 2 reports as per specified.	6	
	Total Marks	**100**	